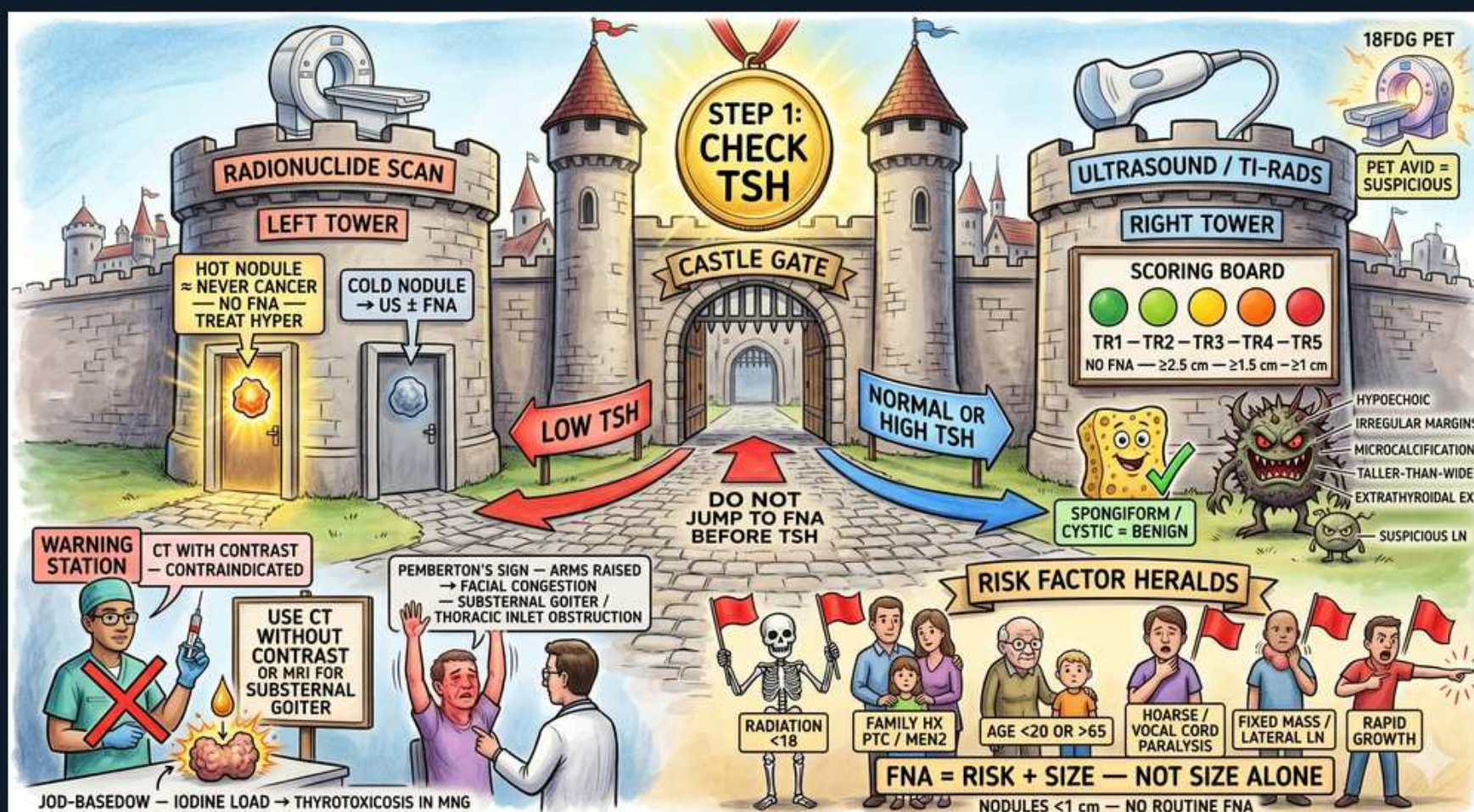


Thyroid — Nodule Workup & TI-RADS



1. STEP 1: Check TSH

WHAT YOU SEE

Castle banner 'STEP 1: CHECK TSH'

WHAT IT ENCODES

Every thyroid nodule workup begins with serum TSH — it is the single test that branches the entire algorithm. TSH plus a dedicated thyroid ultrasound are the only mandatory first steps. Calcitonin is checked only if medullary carcinoma/MEN2 is suspected; thyroglobulin has no role in initial diagnosis.

BOARD TRAP

Jumping straight to FNA or imaging without a TSH — you cannot decide between an uptake scan and US/FNA without it.

2. Do NOT FNA before TSH

WHAT YOU SEE

'DO NOT JUMP TO FNA BEFORE TSH'

WHAT IT ENCODES

TSH first because a LOW TSH redirects you to a radionuclide uptake scan (a hot nodule is essentially never cancer and should not be biopsied), whereas a normal/high TSH sends you to ultrasound and possible FNA. Biopsying first wastes a procedure and can yield indeterminate cytology on what is actually a benign autonomous nodule.

BOARD TRAP

FNA before checking TSH — risks an unnecessary biopsy of a hot (benign autonomous) nodule.

3. Low TSH → radionuclide scan

WHAT YOU SEE

The LEFT TOWER labelled 'RADIONUCLIDE SCAN', reached by the 'LOW TSH' path — a suppressed TSH routes you to the uptake-scan tower to map function.

WHAT IT ENCODES

Low/suppressed TSH suggests an autonomously functioning nodule, so the next step is a radioiodine (I-123) or technetium uptake scan to map function. A focal 'hot' area that suppresses the rest of the gland is a toxic adenoma; multiple hot areas = toxic MNG.

4. Hot nodule = never cancer

WHAT YOU SEE

'HOT NODULE = NEVER CANCER, NO FNA, treat hyper'

WHAT IT ENCODES

A hyperfunctioning ('hot') nodule is malignant in well under 1% of cases — effectively never cancer — so it does NOT need FNA. Manage the hyperthyroidism instead (RAI ablation or surgery; thionamides as a bridge). Only cold/hypofunctioning nodules carry meaningful cancer risk and warrant cytologic evaluation.

BOARD TRAP

Performing FNA of a hot nodule — unnecessary; treat the hyperthyroidism rather than biopsy.

5. Cold nodule → US ± FNA

WHAT YOU SEE

'COLD NODULE → US ± FNA'

WHAT IT ENCODES

A cold (non-functioning) nodule on uptake scan accounts for ~5–15% malignancy risk, so it is characterized by ultrasound and biopsied per sonographic risk + size. Note: most nodules in the population are cold, and most cold nodules are still benign — risk stratification by US (TI-RADS) prevents over-biopsy.

6. Normal/high TSH → US/TI-RADS

WHAT YOU SEE

'NORMAL OR HIGH TSH → ULTRASOUND/TI-RADS'

WHAT IT ENCODES

A normal or high TSH means the nodule is not autonomous — a uptake scan is pointless (it would not be hot), so go directly to diagnostic ultrasound and apply TI-RADS risk stratification to decide on FNA. A high TSH also raises suspicion for underlying Hashimoto's.

BOARD TRAP

Ordering a radionuclide uptake scan when TSH is normal/high — wasteful; the nodule won't be hot, so US is the correct next test.

7. TI-RADS scoring + size cutoffs

WHAT YOU SEE

A SCORING BOARD in the right tower listing TR1→TR5 with falling size cutoffs (no FNA / ≥2.5 / ≥1.5 / ≥1 cm) — the riskier the category, the smaller the size that triggers FNA.

WHAT IT ENCODES

ACR TI-RADS assigns points for composition, echogenicity, shape, margin, and echogenic foci to give TR1 (benign) through TR5 (highly suspicious). FNA thresholds: TR1–2 no FNA; TR3 FNA if ≥2.5 cm (follow ≥1.5 cm); TR4 FNA if ≥1.5 cm (follow ≥1 cm); TR5 FNA if ≥1 cm (follow ≥0.5 cm). Higher suspicion → smaller size triggers biopsy.

BOARD TRAP

Memorizing one universal size cutoff — the FNA threshold falls as TI-RADS category rises, so size alone is never the rule.

8. FNA = risk + size, not size alone

WHAT YOU SEE

'FNA = RISK + SIZE — NOT SIZE ALONE; <1cm no routine FNA'

WHAT IT ENCODES

The FNA decision integrates sonographic suspicion AND size: a small but highly suspicious (TR5) nodule may be biopsied at ≥1 cm, while a large spongiform (TR1–2) nodule may never be. Subcentimeter nodules (<1 cm) are generally not routinely aspirated unless there are clinical red flags (suspicious nodes, extrathyroidal extension).

BOARD TRAP

Biopsying every nodule ≥1 cm regardless of features — over-biopsies benign nodules; conversely don't ignore a <1 cm TR5 nodule with suspicious nodes.

9. Spongiform/cystic = benign

WHAT YOU SEE

'SPONGIFORM/CYSTIC = BENIGN'

WHAT IT ENCODES

A spongiform pattern (aggregation of microcystic spaces in >50% of the nodule) or a purely cystic/anechoic nodule is essentially always benign (TR1) and does not require FNA. These reassuring features are high-yield for distinguishing benign colloid nodules from solid hypoechoic cancers.

10. High-risk US features

WHAT YOU SEE

'Hypoechoic, irregular margins, microcalcifications, taller-than-wide, ETE, suspicious LN'

WHAT IT ENCODES

The classic malignant sonographic signs (each adds TI-RADS points): marked hypoechogenicity, irregular/lobulated/spiculated margins, microcalcifications (punctate echogenic foci — most specific, suggest psammoma bodies of papillary cancer), taller-than-wide shape on transverse view, extrathyroidal extension, and abnormal/suspicious cervical lymph nodes.

BOARD TRAP

Calling coarse/macroclicifications or comet-tail artifact 'suspicious' — those are lower-risk; microcalcifications are the worrisome finding.

11. Pemberton sign

WHAT YOU SEE

'PEMBERTON SIGN — arms raised → facial congestion, substernal goiter'

WHAT IT ENCODES

Pemberton's sign: raising both arms above the head causes facial plethora/congestion, cyanosis, distended neck veins, and stridor/dyspnea — produced by a large substernal goiter obstructing the thoracic inlet ('cork in a bottle'). It signals retrosternal extension and the need for non-contrast CT to map the goiter and assess airway compression.

12. CT contrast contraindicated

WHAT YOU SEE

A red-circle WARNING STATION crossing out 'CT WITH CONTRAST', pointing instead to 'USE CT WITHOUT CONTRAST' with 'JOD-BASEDOW / IODINE LOAD' below — contrast iodine can ignite hyperthyroidism and block RAI, so image substernal goiter without contrast.

WHAT IT ENCODES

Iodinated IV contrast delivers a large iodine load that can trigger Jod-Basedow (iodine-induced hyperthyroidism/storm) in an autonomous nodular goiter AND saturates the gland, blocking any planned radioiodine scan or RAI therapy for weeks to months. Image a substernal goiter with NON-contrast CT.

BOARD TRAP

Ordering contrast-enhanced CT in autonomous/nodular goiter — risks Jod-Basedow and delays RAI/uptake imaging by saturating the gland.

13. Risk-factor red flags

WHAT YOU SEE

A row of 'RISK FACTOR HERALDS' figures each waving a red flag — childhood radiation, family PTC/MEN2, age <20 or >65, hoarseness/vocal-cord palsy, fixed lateral mass, rapid growth — each red flag lowers the threshold to biopsy.

WHAT IT ENCODES

Clinical malignancy red flags lowering the threshold to biopsy/refer: childhood head-neck radiation, family history of thyroid cancer or MEN2/MTC, extremes of age (<20 or >65, esp. male), hoarseness/vocal-cord paralysis (recurrent laryngeal nerve invasion), a firm/fixed nodule, fixed lateral neck mass/adenopathy, and rapid growth (rapid growth in an older patient also raises anaplastic carcinoma or lymphoma).

14. PET-avid nodule

WHAT YOU SEE

'¹⁸FDG PET — PET AVID = SUSPICIOUS'

WHAT IT ENCODES

An incidentally discovered FOCAL FDG-PET-avid thyroid nodule carries a substantial malignancy risk (~30–35%) and should be evaluated with ultrasound and FNA. By contrast, DIFFUSE FDG uptake throughout the gland usually reflects chronic (Hashimoto's) thyroiditis and is not concerning.

BOARD TRAP

Treating diffuse PET uptake the same as focal — diffuse uptake is thyroiditis; only focal uptake demands FNA.